…omputer and Emerging Sciences, Lahore Campus

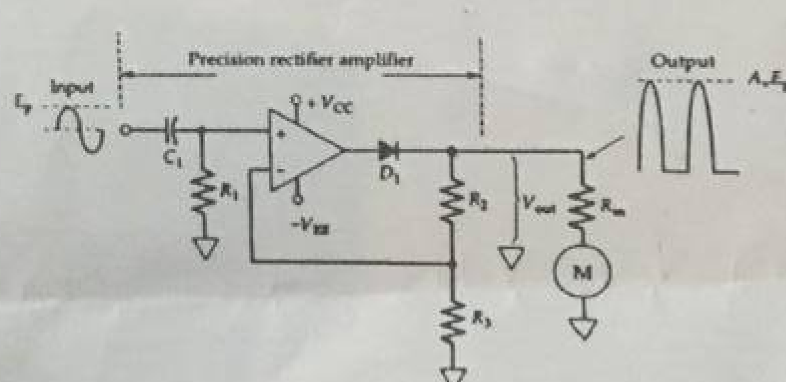
Course:	Instrumentation & Measurements	Course Code:	EE220
Program:	BS (Electrical Engineering)	Semester:	Fall 2017
Duration:	60 Minutes	Total Marks:	30
Paper Date:	4-Nov-17	Weight	15%
Section:	EE	Page(s):	1
Exam:	Midterm-2	Roll No:	L14-4435

Instruction/Notes: Solve All Questions

Q.1a A 6-bit R-2R DAC has $R_F = R = 4\ k\Omega$ and $V_{ref} = 8\ V$. Determine the lowest current level in the circuit, and calculate the output voltage for MSB. Assume ideal OpAmp. [5]

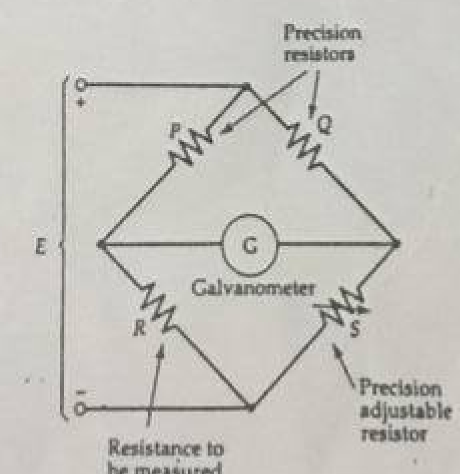
Q.1b A 64 mV level is to be converted into an 10-bit code. Calculate the resolution of the conversion, and analog levels represented by the MSB and LSB, and determine the analog level represented by 1111111110. [5]

Q.2a The ac electronic voltmeter as shown in the following circuit uses the following components: $R_1 = 22\ k\Omega$, $R_2 = 2.25\ k\Omega$, $R_3 = 6.8\ k\Omega$, and $R_S + R_m = 1\ k\Omega$. The meter gives FSD for 300 μA. Calculate the rms input voltage for meter FSD and for 0.5 FSD. [5]



Q.2b A frequency meter measuring the ratio of two frequencies display 1212 when the pulses of the unknown frequency (f_2) are counted over 1000 cycles of the known frequency (f_1). If f_1 is 33.35 kHz, determine f_2. [5]

Q.3a A Wheatstone bridge as shown below has $P = 1\ k\Omega$, $Q = (100\ \Omega, 1\ k\Omega, \text{or } 100\ k\Omega)$, and S is adjustable from 1 kΩ to 5 kΩ. Determine the range of resistance that can be measured. [5]



3b When a 10 kV supply is used for measuring insulation resistance of a metal-sheathed electrical cable, the measured current is 4.3 μA without the use of the guard wire. When a guard wire is used , the current is 3.3 μA, Calculate the combined volume and surface leakage resistance of the cable. Also, determine the separate volume and surface leakage resistances. [5]

National University of Computer and Emerging Sciences, Lahore Campus

Course:	Instrumentation & Measurements	Course Code:	EE220
Program:	BS (Electrical Engineering)	Semester:	Spring 2018
Duration:	3 Hours	Total Marks:	100
Paper Date:	21-May-2018	Weight	50%
Section:	EE	Page(s):	2
Exam:	Final	Roll No:	

Instruction/Notes: Solve All Questions

Q.1a A 300 Ω, 75 cm length of nickel wire which is bounded to a steel girder has its resistance change by 0.9 Ω when the girder is under compression. If the wire has a gauge factor of 3, calculate the change in the wire length. [CLO 6] [10]

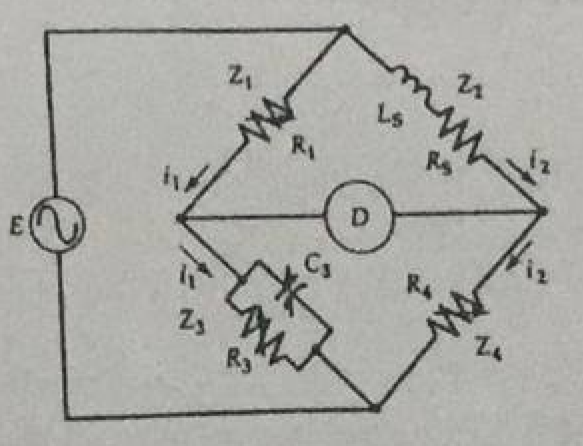
Q.1b When its total air gap is 0.33 mm, a variable reluctance transducer has 0.35 mH inductance. The inductance decreases to 0.3 mH when the target is moved further from the core. Calculate the target movement. [CLO 6] [10]

Q.2a The distance between plates of a capacitive transducer increases from 0.59 mm to 0.82 mm. Determine the corresponding percentage change in capacitance.

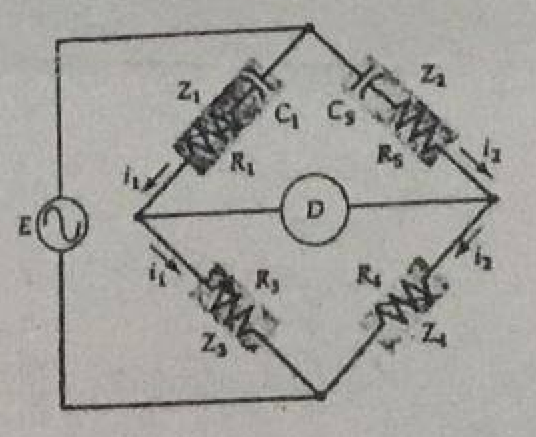
$C - \Delta C = \frac{\varepsilon_0 \varepsilon_r A}{D + \Delta d}$ [CLO 6] [10]

Q.2b A resistance thermometer has a 100 Ω resistance and a temperature coefficient 0f 0.0037 at 20 °C. Calculate its resistance at 75 °C, and determine the temperature when its resistance is 150 Ω. [CLO 6] [10]

Q.3a A Maxwell bridge with a 10 kHz supply frequency uses a 0.1 μF capacitor for C_3 and a 100 Ω standard resistor for R_1. Resistors R_3 and R_4 can be adjusted from 100 Ω to 1 KΩ kΩ. Calculate the range of inductances and Q factors that can be measured on the bridge. [PLO 1][CLO 5] [10]



Q.3b When investigating an unknown capacitor, a series RC bridge as shown below has the following components at balance: C_1 = 0.5 μF, R_1 = 5.5 Ω, R_3 = 1.25 kΩ, and R_4 = 2.7 kΩ. If a 3.9 Ω resistor is included in series with Cs, determine the capacitance Cs and series-resistive component of Cs. [PLO 1][CLO 5] [10]



National University of Computer and Emerging Sciences, Lahore Campus

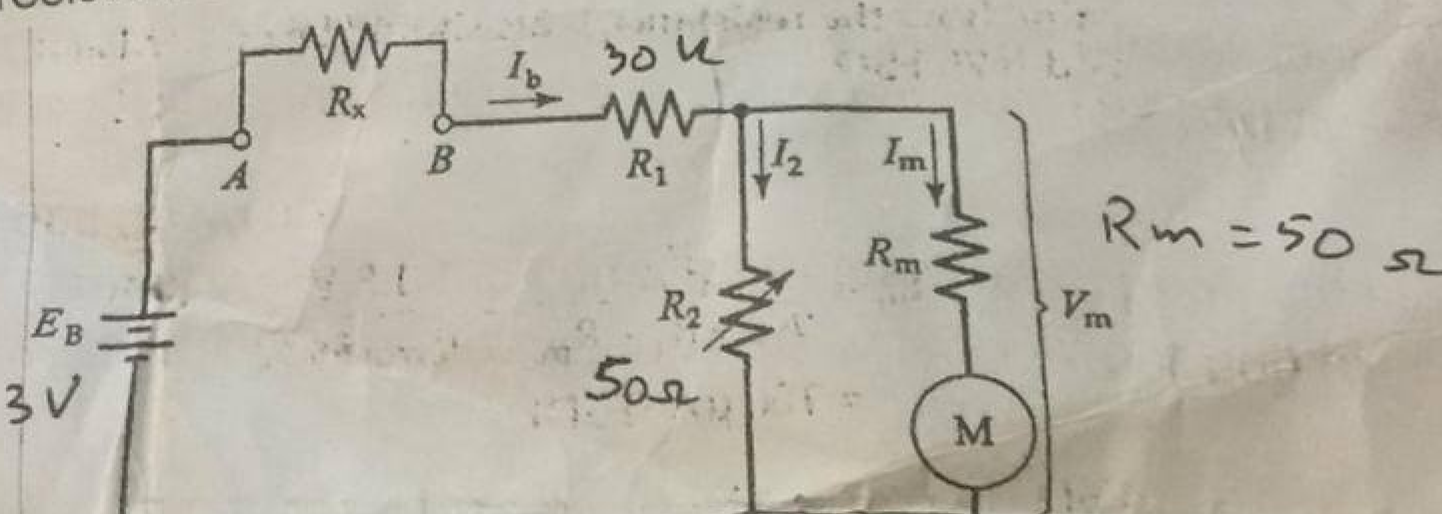
Course:	Instrumentation & Measurements	Course Code:	EE 220
Program:	BS (Electrical Engineering)	Semester:	Fall 2017
Duration:	60 Minutes	Total Marks:	30
Paper Date:	20-Sep-17	Weight	15%
Section:	EE	Page(s):	1
Exam:	Midterm-1	Roll No:	L14-4435

Instruction/Notes: Solve All Questions. Attempt all parts of a question together.

Q.1 a. Two series connected resistors are specified as $R_1 = 1.8\ k\Omega \pm 5\%$ and $R_2 = 4.7\ k\Omega \pm 10\%$ at 25 °C. If each has a 200 ppm/°C temperature coefficient, calculate the maximum and minimum resistance of the two at 60°C. [5]

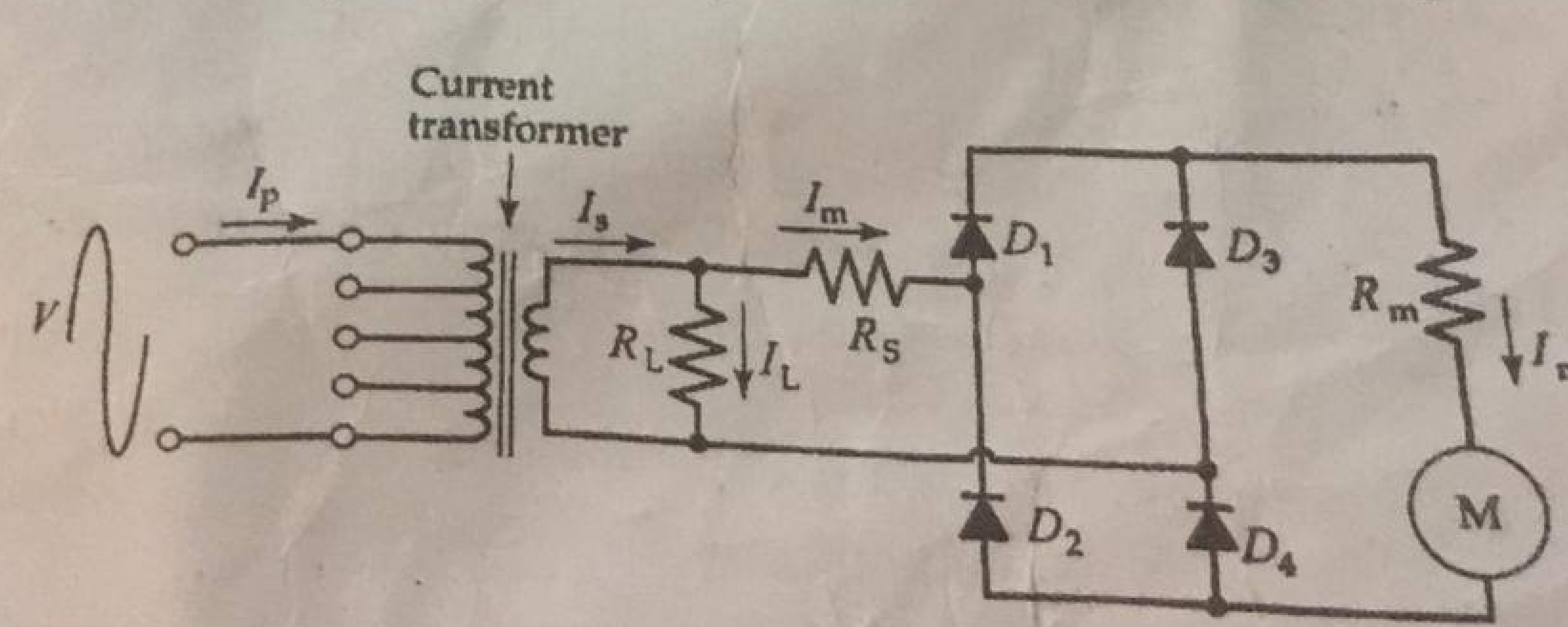
b. A 10 kΩ potentiometer with a 20 Ω resolution is used as a voltage divider. If the potentiometer supply is 12 V, determine the precision of the output voltage. [5]

Q.2 a. An ohmmeter is made up of the following components: supply voltage $E_B = 3$ V, series resistor $R_1 = 30$ KΩ, meter shunt resistor $R_2 = 50\ \Omega$, and meter resistance $R_m = 50\ \Omega$. Determine the resistance measured at 0.25, 0.5, and 0.75 FSD. [5]



b. A PMMC instrument with a 750 Ω coil resistance gives FSD with a 500 μA coil cu
Determine the required shunt resistance to convert the instrument into a dc ammet
an FSD of (a) 50 mA and (b) 30 mA. [5

Q.3 A rectifier ammeter is to indicate full scale for a 1 A rms current. The PMMC instru
used has 500 μA FSD and a 1200 Ω coil resistance. The current transformer turns
= 7000 and $N_P = 10$. Silicon diodes are used and the meter series resistance is R_S
kΩ. Determine the required secondary shunt resistance value R_L.



National University of Computer and Emerging Sciences, Lahore Campus

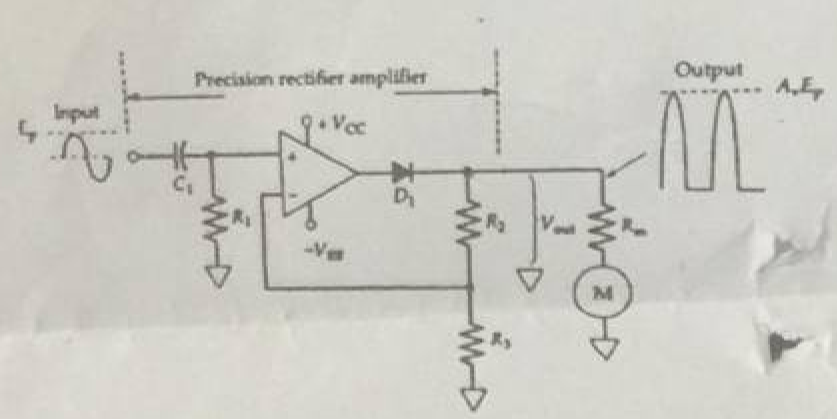
Course:	Instrumentation & Measurements	Course Code:	EE220
Program:	BS (Electrical Engineering)	Semester:	Fall 2017
Duration:	60 Minutes	Total Marks:	30
Paper Date:	4-Nov-17	Weight	15%
Section:	EE	Page(s):	1
Exam:	Midterm-2	Roll No:	L14-4435

Instruction/Notes: Solve All Questions

Q.1a A 6-bit R-2R DAC has $R_F = R = 4\ k\Omega$ and $V_{ref} = 8$ V. Determine the lowest current level in the circuit, and calculate the output voltage for MSB. Assume ideal OpAmp. [5]

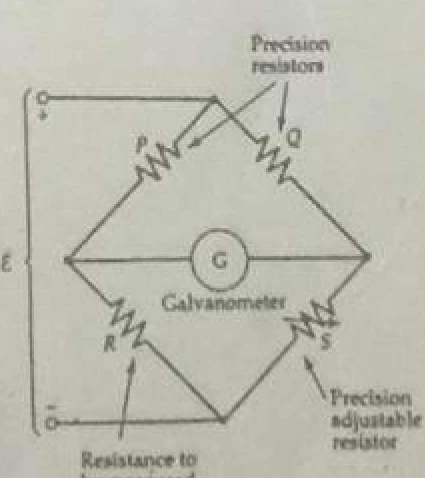
Q.1b A 64 mV level is to be converted into an 10-bit code. Calculate the resolution of the conversion, and analog levels represented by the MSB and LSB, and determine the analog level represented by 1111111110. [5]

Q.2a The ac electronic voltmeter as shown in the following circuit uses the following components: $R_1 = 22\ k\Omega$, $R_2 = 2.25\ k\Omega$, $R_3 = 6.8\ k\Omega$, and $R_S + R_m = 1\ k\Omega$. The meter gives FSD for 300 μA. Calculate the rms input voltage for meter FSD and for 0.5 FSD. [5]



Q.2b A frequency meter measuring the ratio of two frequencies display 1212 when the pulses of the unknown frequency (f_2) are counted over 1000 cycles of the known frequency (f_1). If f_1 is 33.35 kHz, determine f_2. [5]

Q.3a A Wheatstone bridge as shown below has P = 1 kΩ, Q = (100 Ω, 1 kΩ, or 100 kΩ), and S is adjustable from 1 kΩ to 5 kΩ. Determine the range of resistance that can be measured. [5]



Q.3b When a 10 kV supply is used for measuring insulation resistance of a metal-sheathed electrical cable, the measured current is 4.3 μA without the use of the guard wire. When a guard wire is used , the current is 3.3 μA, Calculate the combined volume and surface leakage resistance of the cable. Also, determine the separate volume and surface leakage resistances. [5]

3

National University of Computer and Emerging Sciences, Lahore Campus
Quiz 1, Instrumentation & Measurements EE, Fall 2017, Sep 7, 2017
Max Marks = 20, Time Allowed 15 minutes

Name: Hadia Hassan Roll #: L14.4435

Q.1 Three resistors connected in series have the following values:
(5.6 kΩ ±5%), (8.2 kΩ ±6%), (6.8 kΩ ±5%).
Calculate the total resistance, the maximum absolute error, and the maximum relative error.

4

$R_T = R_1 + R_2 + R_3 = (5.6k\Omega \pm 5\%) + (8.2k\Omega \pm 6\%) + (6.8k\Omega \pm 5\%)$

$R_1 = 5.6k\Omega \pm 0.28$, $R_2 = 8.2k\Omega \pm 0.492$

$R_3 = 6.8k\Omega \pm 0.34$

$R_T = (5.6k\Omega + 8.2k\Omega + 6.8k\Omega) \pm (0.28 + 0.492 + 0.34)$

$\boxed{R = 20.6k\Omega \pm 1.112}$

$\boxed{R = 20.6 \pm (16\%)}$

Relative Resistance.

$R_1 = \frac{5.88}{5.6} \times 100 = 105k\Omega$, $R_2 = \frac{8.692}{8.2} \times 100 = 106k$, $R_3 = 105k\Omega$

Relative Resistance $= 105k\Omega + 106k\Omega + 105k\Omega = 316k\Omega$

Q.2 A dc ammeter is constructed of 5.01 Ω resistance in parallel with a PMMC instrument. If the instrument has a 2.0 kΩ coil resistance and 100 µA FSD, determined measured current at FSD.

$R_m = 5.01\Omega$

$V_m = I_m R_m$

$V_m = \frac{2k\Omega}{\cancel{5.01}} \times 100\mu A = \cancel{0.501 \times 10}$ $\cancel{0.501}$ A

$I_m = 2000$ mA

?

Measured Current $= \frac{V_m}{R_m} - I$

$=$

Q.1 A digital frequency meter displays a count of 166. The time base has a 2 MHz clock generator frequency-divided by four decade counters and one flip-flop. Calculate the gate time and the measured frequency. Determine the count that would be displayed if the flip-flop is removed from the time base.

Q.2 The LSB for an analog voltage converted in to a 16-bit digital code is 1 μV. Calculate the analog equivalent for the MSB and for full scale.

$V_m = LSB \times 2^n = 1\,\mu V \times 2^{16} = 0.0655\,V$

$MSB = \frac{V_m}{2} = \frac{0.0655}{2} = 0.03275\,V$

$MSB = 32.768$

$MSB = 2^{n-1} \times LSB = 2^{15} \times 1\,\mu V = 0.0327\,V$

$V_{FSD} = V_M - LSB = 0.0655 - 1\times10^{-6} = 0.0654\,V$

$65.535\,mV$

Q.3 A frequency meter with an accuracy of ±1 LSD ± (1 x 10^{-5}) is used to measure frequencies of 30 Hz, 30 MHz, and 300 MHz. Calculate percentage error in each measurement.

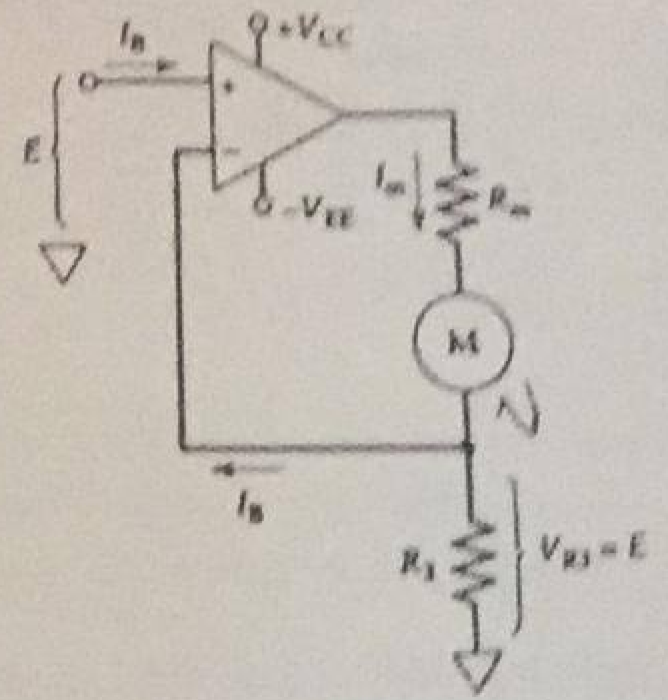
$31 + 1\times10^{-5} = 31.00001$

$31 - 1\times10^{-5} = 30.9999$

$\Delta f = 31.00001 - 30.9999 = 2\times10^{-5}$

$= \frac{\Delta f \times 30}{100} = 6\times10^{-4}$

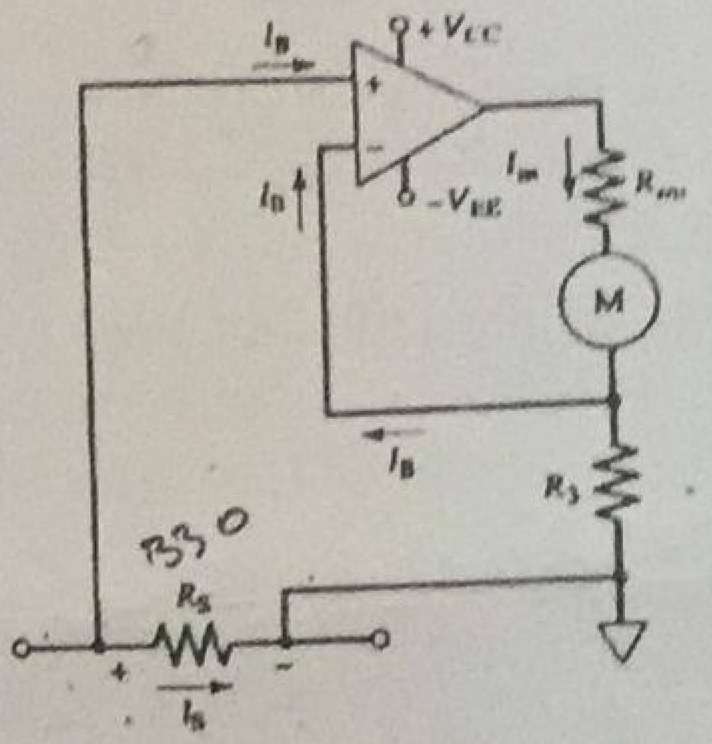
Q.4a The following voltage-to-current converter circuit uses 37.5 μA (FSD) deflection meter with $R_m = 900\ \Omega$. If $R_3 = 80\ k\Omega$, determine the input voltage level to give FSD and 0.5 FSD. [PLO 1] [CLO 3] [10]



$I_m = I_B + I_{R3}$

$I_m = I_B + \frac{37.5\mu}{80K}$

Q.4b A 200 μA current is to be measured by the type of the circuit shown below. The shunt resistor is 330 Ω, and 1 mA (FSD) PMMC meter with 900 Ω coil resistance is to be used. Determine a suitable resistance for R_3 to give FSD on the meter and calculate the maximum output voltage of the operational amplifier. [PLO 1] [CLO 3] [10]



$\frac{V_m - V_{R3}}{R_3} = I_{R3}$

$V = IR$

$V = R?$

Q.5a A dc ammeter is constructed of a 133.3 Ω resistance in parallel with a PMMC instrument. If the instrument has a 1.2 kΩ coil resistance and 30 μA FSD, determine the measured current at FSD, 0.5 FSD, and 0.33 FSD. [CLO 2] [10]

Q.5b Two resistors $R_1 = 70\ k\Omega$ and $R_2 = 50\ k\Omega$, are connected in series across a 12 V supply. A voltmeter on a 5 V range is connected to measure the voltage across R_2. Calculate V_{R2} (a) with the voltmeter disconnected, (b) with a voltmeter having a sensitivity of 20 kΩ/V, and (c) with a voltmeter that has a sensitivity of 200 kΩ/V. [CLO 2] [10]

5

$12V \times 20k\Omega$